Predicting 30-day Hospital Readmission for Diabetic Patients

By: Jose Fernando A. Gonzales

Objective

- To develop and compare **binary classification models** (Logistic Regression, Decision Trees, Random Forest, and XGBoost) that can accurately predict 30-day readmission risk.

Outcome

- This project developed a machine learning pipeline that utilizes a **Random Forest model trained on PCA-reduced features** (44 components) to predict 30-day hospital readmissions for diabetic patients.

Problem Statement

- **Clinical Impact:** Unplanned hospital readmissions for diabetic patients cost the U.S. healthcare system a substantial clinical and financial burden and serve as a critical indicator of care quality.
- **High-Risk Population:** Diabetic patients have readmission rates between **14.4% and 22.7%**, which is **significantly highe**r than the general hospitalized population (8.5–13.5%).
- **Operational Priority:** In a clinical setting, **false negatives** (failing to identify a high-risk patient) are considered more costly and harmful than false positives, driving a need for recall-oriented models.

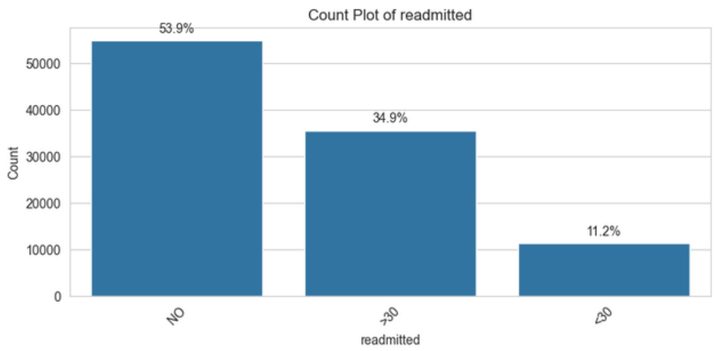
Data Overview

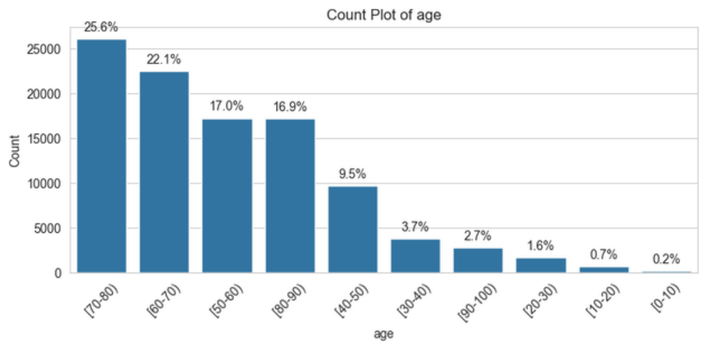
- **Source:** 10 years (1999–2008) of clinical records from **130 U.S. hospitals.**
- **Volume: 101,766 encounters** with **50 raw features** including demographics, admission context, encounter intensity, and medication history.
- **Challenge:** The dataset exhibits **strong class imbalance**, with the positive class (readmission within 30 days) representing only 11.2% of the population.
- **Predictive Drivers:** Analysis identifies **prior healthcare utilization** (emergency and inpatient visits) as the strongest signals for future readmission risk.

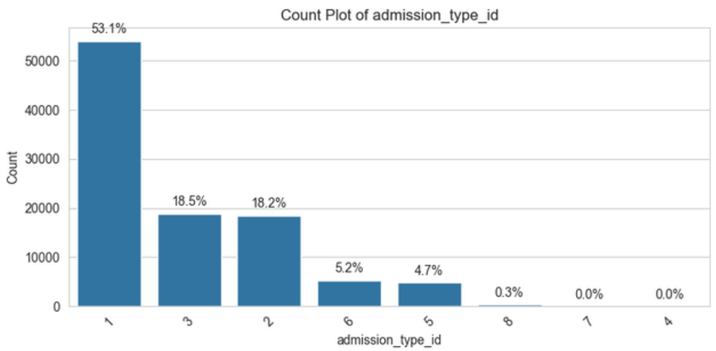
Raw Feature Families

Family	Representative fields	Role in analysis
Identifiers	encounter_id, patient_nbr	Administrative keys only; dropped before modeling
Demographics	race, gender, age, weight	Population profiling, fairness analysis, and risk segmentation
Admission and discharge context	admission_type_id, discharge_disposition_id, admission_source_id, medical_specialty, payer_code, time_in_hospital	Captures care pathway, access, and severity proxies
Encounter intensity	num_lab_procedures, num_procedures, num_medications, number_diagnoses	Measures complexity of the hospital stay
Prior utilization	number_outpatient, number_emergency, number_inpatient	Strongest historical risk signals
Clinical, test, and medication fields	diag_1, diag_2, diag_3, max_glu_serum, A1Cresult, change, diabetesMed, and medication-status columns	Captures diagnoses, testing, treatment patterns, and medication adjustments

EDA Key Findings







Significant Class Imbalance: The target variable is heavily skewed, with only 11.16% of encounters resulting in a 30-day readmission. This imbalance necessitates a focus on recall rather than raw accuracy.

Older Demographic: Approximately 67% of patients are aged 60 or older, which is clinically consistent with the higher prevalence of Type 2 diabetes and associated comorbidities.

Care Setting: Emergency admissions dominate the cohort, representing roughly 53% of all cases.

Top Predictive Drivers (Bivariate Analysis)

Prior Healthcare Utilization: This is the strongest signal identified. Patients readmitted within 30 days had a **mean prior inpatient visit count 117.9% higher** and a **mean emergency visit count 100.6% higher** than those not readmitted.

Discharge Disposition: Discharge destination is a critical risk factor. Patients transferred to **skilled nursing facilities (SNFs) or rehab settings** show significantly higher readmission rates than those discharged to home.

Care Complexity: Longer hospital stays and higher numbers of medications/diagnoses show weak but consistent positive correlations with readmission risk.

Data Quality Issues

Issue	Evidence	Treatment
? used as missing marker	Present in multiple fields rather than explicit NaN	Re-coded during preprocessing
Very high missingness in weight	Roughly 97% missing	Dropped
High missingness in payer_code	Roughly 40% missing and weak clinical value	Dropped
High missingness in medical_specialty	Roughly 49% missing	Retained, grouped into Other and Unknown
Low missingness in race	About 2.23% missing	Retained, recoded to Unknown
Constant medication columns	examide, citoglipton	Dropped
Structurally impossible readmissions	Expired and hospice-related discharge codes	Removed before modeling

Leakage

Leakage Filtering: Analysis identified **1,671 encounters** (e.g., patients who expired or entered hospice) where readmission was structurally impossible (0% rate). These were removed to prevent the model from learning a "trivially correct" but clinically useless negative signal.

Feature Engineering Approach

1 **Clinical Grouping (ICD-9):** High-cardinality diagnosis codes (~750 unique values) were mapped to **19 standardized clinical chapters** (e.g., circulatory, endocrine) to reduce sparsity and improve interpretability.

2 **Administrative Recoding:** Consolidated admission type, discharge disposition, and admission source into clinically coherent buckets (e.g., separating "Home" from "Facility/SNF" transfers).

3 **Utilization Engineering:** Derived **binary presence flags** and **log1p-transformed counts** for emergency, inpatient, and outpatient history to compress extreme right-skew distributions.

4 **Targeted Encoding**

- **Age:** One-hot encoded into 10 bands to capture the **non-monotonic risk gradient** identified in EDA.
- **Medications:** 21 features mapped to a 3-level ordinal scale (0=No, 1=Steady, 2=Adjusted), collapsing "Up" and "Down" dosages because both signaled similar glycemic instability.

5 **Feature Selection (Mutual Information):** Filtered the initial 170 features using **Mutual Information (MI > 0)** to capture non-linear dependencies. This reduced the feature space to **117 informative predictors.**

6 **PCA Dimensionality Reduction:** Compressed the 117 features into **44** Principal Components, capturing **95.2%** of total variance while eliminating multicollinearity.

Model Selection & Methodology

Three-Stage Modeling Strategy

1

Baaseline Comparison

Evaluated four algorithm families - Logistic Regression, Decision Tree, Random Forest, and, XGBoost

2

Hyperparameter Tuning

Performed RandomSearchCV on top-performing ensemble models (RF and XGBoost) to optimize AUC-ROC

3

Track Evaluation

Retraind tuned models on the 44-component PCA track to assess performance in a compressed, noise-reduced feature space

4

Threshold Optimization

Moved beyong the default 0.50 threshold to a **“Constrained” strategy** to **maximize recall** while enforcing a minimum precision floor of 15%

Model Selection & Success Criteria

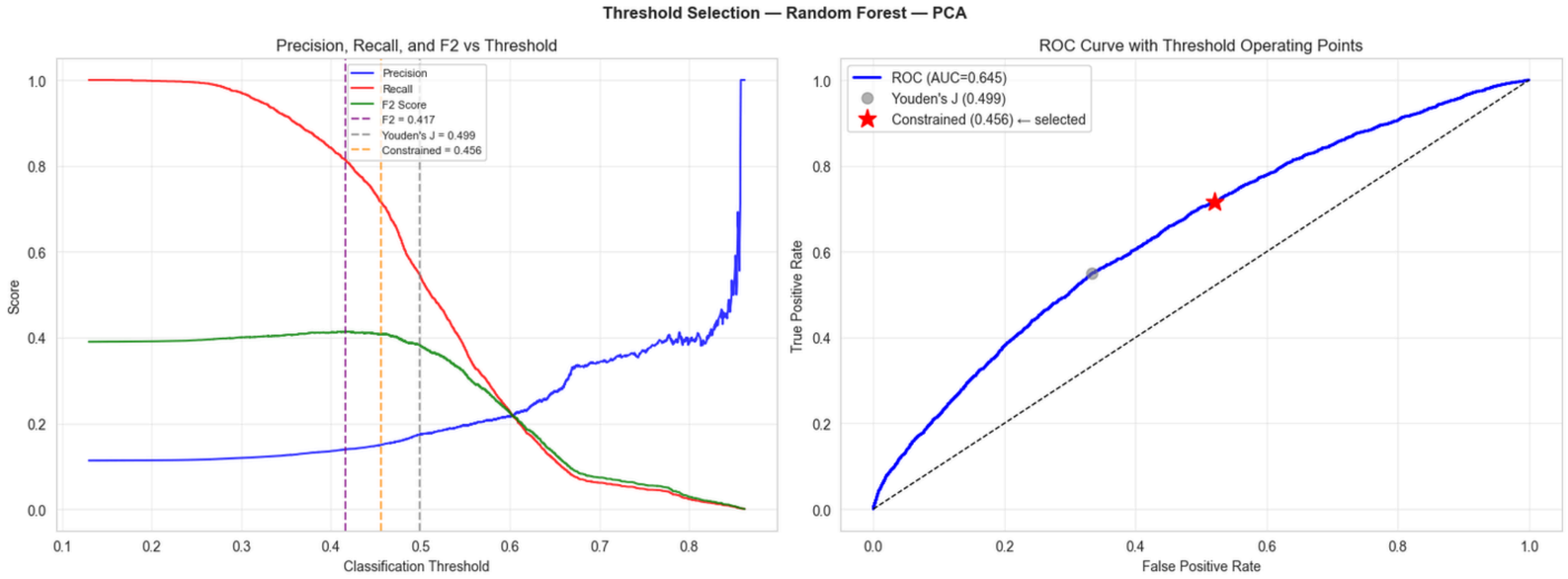
Selection Rule: The final model was selected based on the highest **Test AUC-ROC** across all candidates.

Primary Metric: AUC-ROC was used to measure overall discrimination power between classes.

Secondary Metric: Recall was prioritized due to the clinical cost asymmetry (false negatives are more costly than false positives).

Results & Champion Model Selection

Model Comparison Summary



Model	Feature Track	Test AUC-ROC	Test Recall (Default 0.50)
Random Forest — PCA	**44 Components**	**0.6446**	**54.25%**
Random Forest (tuned)	117 Features	0.6224	23.91%
XGBoost (tuned)	117 Features	0.6213	4.93%
Logistic Regression	117 Features	0.5715	13.69%

The final model was selected based on the highest Test AUC-ROC across all feature tracks and algorithm families.

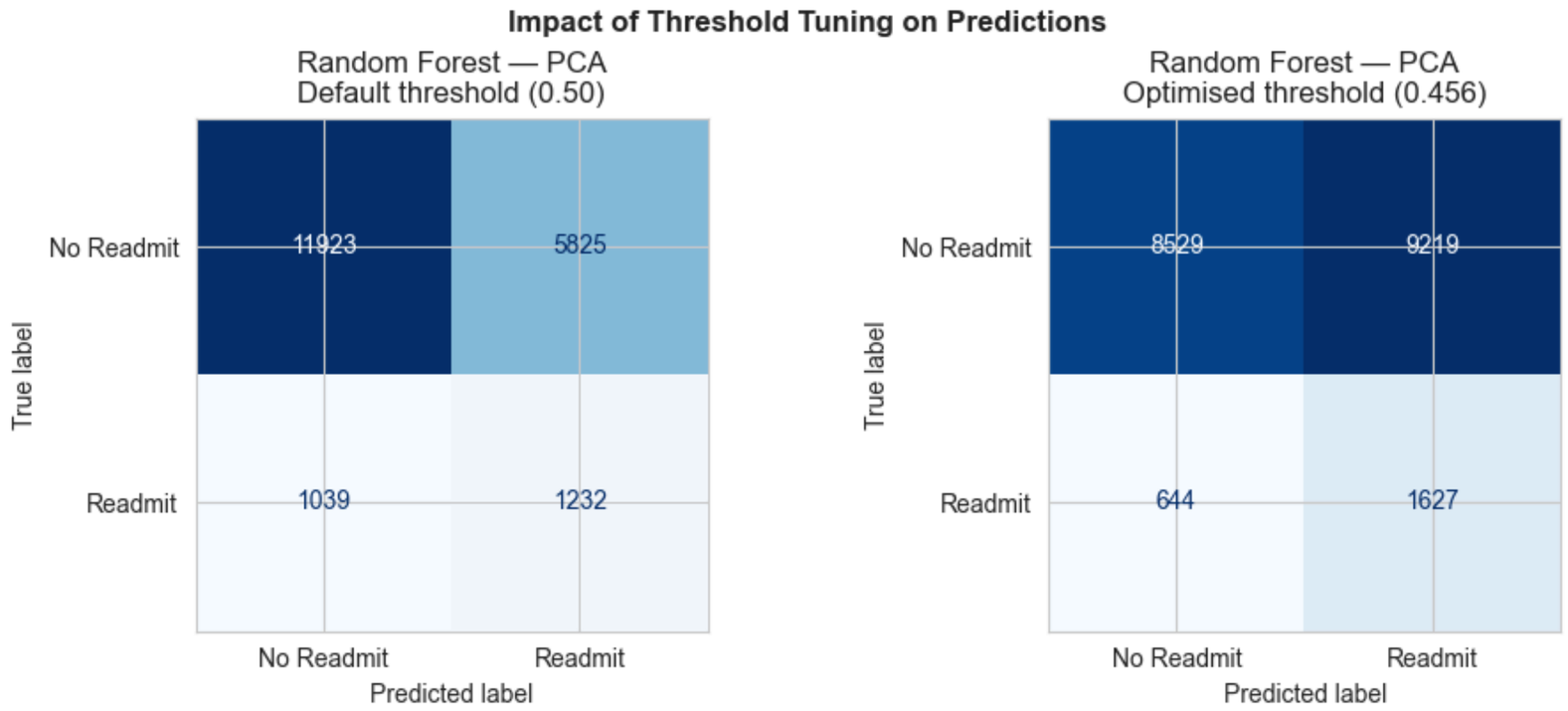
Champion Model: Random Forest - PCA

- **Performance vs. Success Criteria:**
  - **Recall**: Achieved 71.64% (Target: >= 50%) — MET
  - AUC-ROC: Achieved 0.6446 (Target: >=0.75) — NOT MET;
- However, result is consistent with published benchmarks (0.63–0.68) for this difficult dataset.

Results & Champion Model Selection

Key Conclusion

The model is optimized as a **recall-oriented screening aid**. By flagging ~54% of patients, it successfully captures **~72% of all true readmissions**, providing a manageable operational load for hospital triage teams.



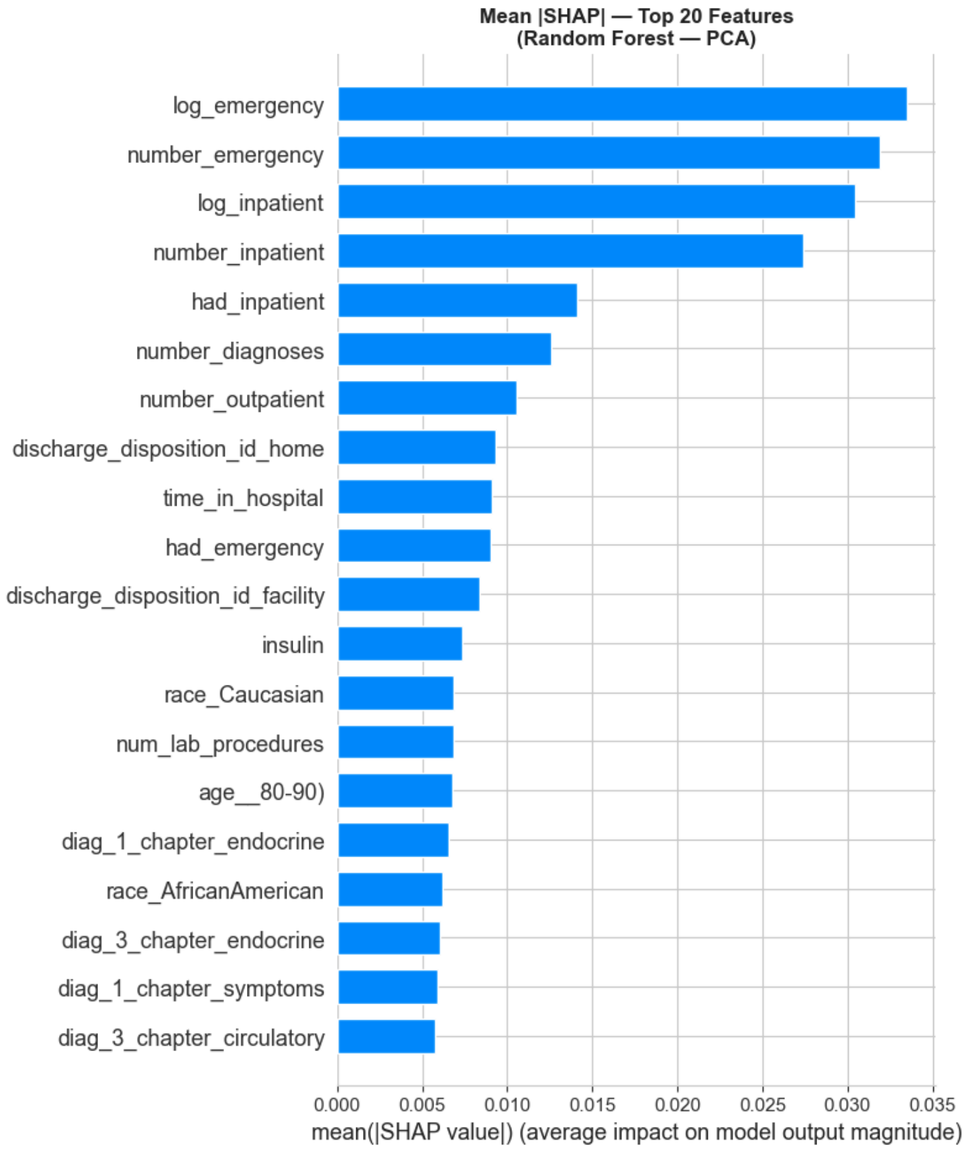
Feature Importance & Explainability

SHAP Explainability

- **SHAP Analysis:** Utilized **SHapley Additive exPlanation**s to decompose individual predictions into additive per-feature contributions grounded in game theory.
- **PCA Back-Projection:** Because the champion model operates on compressed features, SHAP values were first computed on the **44 principal components** and then **back-projected to the original 117 clinical features.**
- **Clinical Relevance:** This technical step ensures that model behavior remains transparent and traceable to the original clinical variables rather than opaque mathematical abstractions.

Top 10 Global Predictive Drivers

1. **log_emergency** (Historical emergency visits)
2. **number_emergency**
3. **log_inpatient** (Historical inpatient stays)
4. **number_inpatient**
5. **had_inpatient** (Binary flag)
6. **number_diagnoses** (Comorbidity burden)
7. **number_outpatient**
8. **discharge_disposition_id_home**
9. **time_in_hospital** (Length of stay)
10. **had_emergency** (Binary flag)



Feature Importance & Explainability

Key Clinical Themes

- **Prior Healthcare Utilization (Primary Signal):** Historical emergency and inpatient encounters dominate importance. Frequent acute-care use reflects **chronic instability and unresolved disease burden,** serving as the strongest predictor of future readmission.
- **Encounter Complexity:** High numbers of diagnoses and longer lengths of stay indicate high comorbidity burden and clinical instability at the point of discharge.
- **Discharge Pathway:** Discharge to home is identified as a significant protective factor, while discharge to facilities (SNFs or rehab) signals higher clinical severity.

Directional Impact of Top Drivers

- **Risk Drivers (↑ Value = ↑ Risk):** log_emergency, number_emergency, log_inpatient, number_inpatient, and had_inpatient.
- **Protective Factors (↑ Value = ↓ Risk):** Discharge to home, Caucasian race, and age cohort [80–90).

Actionable Operational Insight

- **Early Triage:** The model's most powerful signals—prior utilization history—are available in structured EHR data **at the point of admission,** not just at discharge.
- **Strategic Opportunity:** Hospitals can utilize this model to **flag high-risk patients on arrival,** enabling care management teams to begin intensive discharge planning and transitional support well before the patient is ready to leave.

Limitations & Future Work

Current Project Limitations

- **Data Recency & Temporal Signals:** The dataset concludes in **2008**, which may not reflect modern clinical protocols or the latest diabetic treatment standards. Additionally, it lacks longitudinal signals like **medication adherence** and **lab result trends.**
- **Class Imbalance Constraints**: A persistent **8:1 imbalance ratio** made high discrimination difficult, resulting in an **AUC-ROC (0.6446)** that fell short of the primary 0.75 target.
- **Fairness Disparities:** The model exhibits uneven performance across racial subgroups, with a **recall gap of 0.2189** that exceeds the project's 0.15 fairness safeguard.
- **Administrative Noise:** Coding practices can vary by hospital, potentially introducing noise based on **institutional documentation habits** rather than pure clinical risk.

Future Work & Roadmap

- **Advanced Modeling & Calibration:** Evaluate newer algorithms like **LightGBM or CatBoost** and implement **probability calibration** (e.g., Platt scaling) to make risk scores more reliable for clinical stratification.
- **Data Expansion:** Incorporate **post-2008 clinical data** and perform temporal validation to ensure the model generalizes to contemporary healthcare environments.
- **Fairness-Aware Training:** Integrate **demographic parity or equalized-odds constraints directly** into the training process to mitigate racial recall gaps.

Thank you.